

1943-

MILESTONES

PUBLISHES FINDINGS

Observes a pulsar in 1967. Antony Hewish reports her discovery in a *Nature* article the following year.

PHYSICS PRIZE

Although not named in the award herself, her supervisor receives the Nobel Prize in 1974.

OFFICIAL RECOGNITION

Made a Dame Commander of the British Empire (DBE) in 2007 for her services to astronomy.

As a young postgraduate student, Jocelyn Bell Burnell was the first scientist to detect pulsars, opening up a new branch of astrophysics.

Born in Belfast in Northern Ireland, Jocelyn Bell Burnell developed a flair for science at boarding school, where she realized "how easy physics was." In her teenage years, she resolved to work in astronomy.

After graduating from Glasgow University in 1965, Bell Burnell began a PhD thesis on twinkling quasars at Cambridge University. Under the supervision of Antony Hewish, a reader in astrophysics, she helped to design and build an array of radio telescopes that probed deep into space to detect quasars, a recently discovered form of high-energy cosmic radiation.

In November 1967, her instruments detected unexpected signals: radio waves pulsating every 1.337 seconds, coming from a fixed point in space. Similar signals were detected in other parts of space over the following 2 months.

Bell Burnell and Hewish concluded that the radio waves came from the radiation beam of a rapidly rotating, strongly magnetized neutron star. Each rotation emitted a "pulse," giving the name "pulsar" to this entirely new class of star.

The suspiciously precise radio waves led Bell Burnell to playfully apply the working title LGM-1 (Little Green Men-1) until the term "pulsar" was coined in 1968.



JOCELYN
BELL
BURNELL

